

DFIR Report

By AlJannat A. Uthman

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Summary

The digital forensics analysis focused on a disk image investigation using tools such as FTK Imager, Registry Explorer, Xst Reader, Google Maps and Cyberchef. Key findings included system information, installed applications, user communications, partition details and other relevant information. The analysis combined both static examination of registry hives and file metadata as well as dynamic inspection of user emails and application data.

Tools Used

The following are the tools I used for performing analysis on the disk image.

- FTK Imager
- Registry Explorer
- Xst Reader
- Google Map
- Cyberchef

Image Extraction

In this step, I used FTK imager to extract the contents of the image file. I followed the steps below to achieve this aim.

- File – Add Evidence Item
- Select Image File – Next
- Click on browse and locate the image file.
- Click on finish.

Findings

1. What is the administrator's username?

Opening up the SAM hive in Registry Explorer and navigating to:

SAM\Domain\Account\Users\, a list of users is seen on the right side of the panel. Inspecting the Username field of the Admin account reveals that the administrator's name is **Administrator**.

Last ...	Expir ...	Us...	Full N...	Pass...	Groups	Com...	User ...	Ho
=	=	ABC	ABC	ABC	ABC	ABC	ABC	ABC
		Ad User Name strator			Admini strator s	Built-in account t for adminis tering the comput er/dom ain		

2. What is the OS's build number?

Device related info is located in the SOFTWARE hive. To get the build number of the Windows Image, I navigated to SOFTWARE\Microsoft\Windows NT\CurrentVersion. On the right side of the panel, we can see the CurrentBuildNumber field with the value **16299**.

The screenshot shows the Registry Explorer v2.1.0 interface. The left pane displays the tree structure of the registry, with the path `SOFTWARE\Microsoft\Windows NT\CurrentVersion` expanded. The right pane shows the list of values for this path. The `CurrentBuildNumber` value is highlighted, showing a data type of `RegSz` and a value of `16299`. Below the list, the 'Type viewer' section shows the binary representation of the value.

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated
BuiltGUID	RegSz	ffffffff-ffff-...	00-00		
BuiltLab	RegSz	16299.rs3_rel...			
BuiltLabEx	RegSz	16299.15.amd...			
CompositonEditionID	RegSz	Professional	00-00		
CurrentBuild	RegSz	16299			
CurrentBuildNumber	RegSz	16299			
CurrentMajorVersionNumber	RegDword	10			
CurrentMinorVersionNumber	RegDword	0			
CurrentType	RegSz	Multiprocessor...	65-00-64-00-0...		
CurrentVersion	RegSz	6.3	00-00-00-00		
EditionID	RegSz	Professional	00-00		
EditionSubstring	RegSz				
InstallationType	RegSz	Client	00-00-00-00-0...		
InstallDate	RegDword	1548529801			
ProductName	RegSz	Windows 10 Pro	74-00-65-00-7...		
ReleaseId	RegSz	1709	00-00		

Below the table, the 'Type viewer' section shows the binary representation of the `CurrentBuild` value:

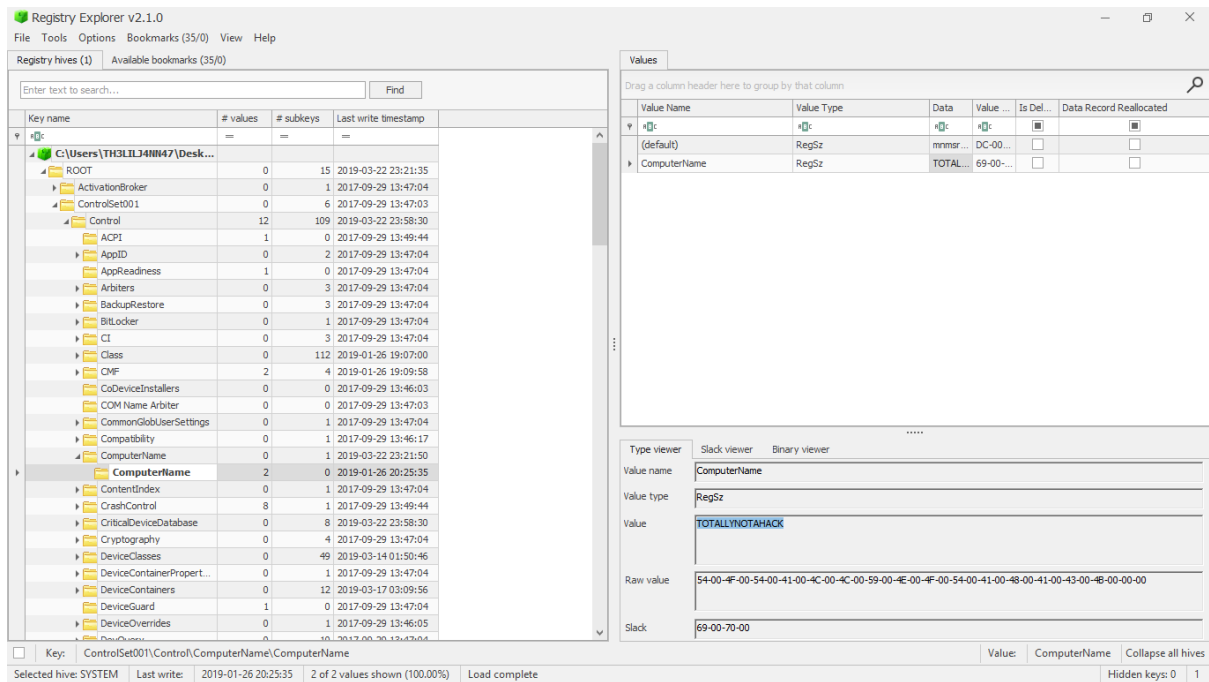
```

Value name: CurrentBuild
Value type: RegSz
Value: 16299
Raw value: 31-00-36-00-32-00-39-00-39-00-00-00
  
```

The status bar at the bottom indicates: Selected hive: SOFTWARE, Last write: 2019-03-22 23:40:50, 27 of 27 values shown (100.00%), Copied Value data to clipboard, Hidden keys: 0 | 1.

3. What is the hostname of the computer?

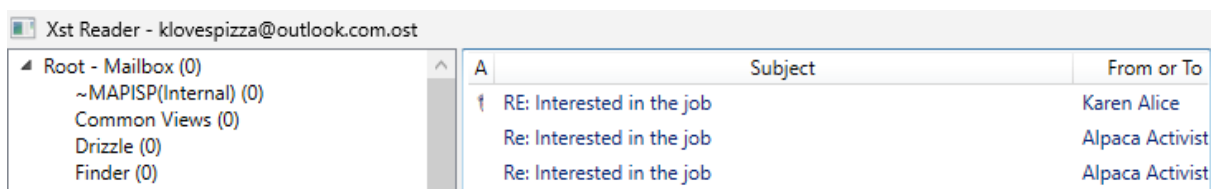
The SYSTEM hive hosts information about the ComputerName. In SYSTEM\ControlSet001\Control\ComputerName\ComputerName, the data field of ComputerName reveals the host name as **TOTALLYNOTAHACK**.



4. A messaging application was used to communicate with a fellow Alpaca enthusiast. What is the name of the software?

In FTK imager, I exported the User folder to a folder saved as Export in my desktop. Inspecting Export\Users\Karen\AppData\Local\Microsoft\Outlook, a file saved as [klovespizza@outlook.com.ost](mailto:klovespizza@outlook.com) is present in the folder.

Using Xst Reader, I gained access to the user's inbox where I found messages shared between an Alpaca enthusiast and the user (Karen). .ost files are offline Outlook files and this tells us the messaging application used for this communication is **Outlook**.



5. What are the initials of the person who contacted the admin user from TAAUSAI?

In one of the chats found in Karen's inbox, MS which stands for Michael Scotch was stated in the concluding part of the message.

We look forward to hearing back from you soon!

Best,
M.S.
--

6. How much money was TAAUSAI willing to pay upfront?

In the follow up mail that MS sent to Karen, he stated the amount TAAUSAI were willing to pay her upfront which was a \$150000.

Hello Ms. Karen,

We are attempting to reach out to you again to see if you'd still be interested in working with us. As we previously mentioned, this is a high paying technical job involving computers. That may sound scary, but all you need to know is how to turn a computer on. We'll provide you with resources about on how to do the rest.

Let us know if you're interested. We're willing to pay \$150,000 USD upfront, and more at the completion of the job.

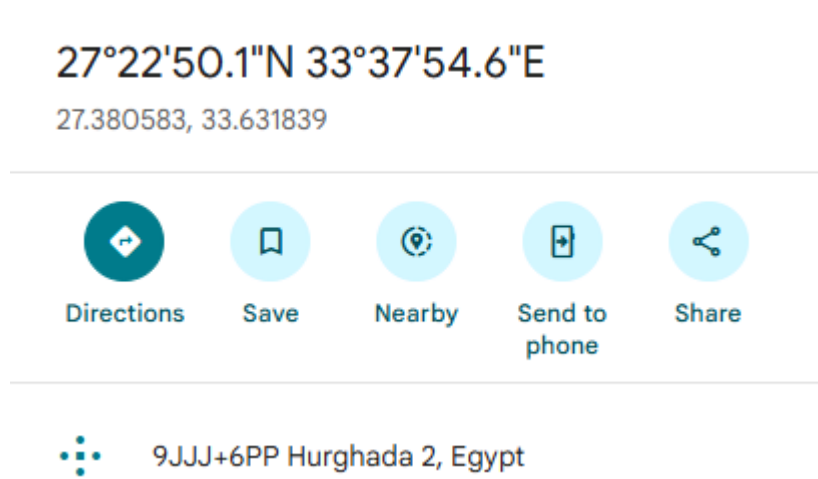
7. What country is the admin user meeting the hacker group in?

MS sent a task to Karen in one of the inbox, he also provided a coordinate (27°22'50.10"N, 33°37'54.62"E) where she could meet the team for further information. The results of the coordinates in Google Map reveals that the meetup point is in **Egypt**.

complaints that he refuses to provide Alpacas with scarfs and beanies during the winter!

What we need you to do is gain his trust and then hack his machine. We will give you more information about this in person. Meet us here "27°22'50.10"N, 33°37'54.62"E"

On Sun, Mar 17, 2019 at 2:48 AM Karen Alice <klovespizza@outlook.com> wrote:



8. What is the machine's timezone? (Use the three-letter abbreviation)

The timezone is **UTC** as seen in the subkey of the SYSTEM hive:

SYSTEM\CurrentControlSet\Control\TimeZoneInformation.

StandardName	@tzres.dll,-932	@tzres.dll,-932
StandardStart	Month 0, week of month 0, day of week 0, Hours:Minutes:Seconds:Milliseconds 0:0:0:0	00-00-00-00-00-00-00-00
TimeZoneKeyName	UTC	UTC
ActiveTimeBias	0	0

9. When was AlpacaCare.docx last accessed?

In the second partition (PacaLady) of the disk image, AlpacaCare.docx is located in the root folder. FTK imager shows that the Date Accessed is **17:03:2019**.

[32216MB]:NONAME	[] \$Secure	56 (1 KB)	Regular File	13/03/2019 4:42:18 AM	13/03/2019 4:42:18 AM	13/03/2019 4:42:18 AM
[3122MB]:PacaLady	[] \$TXF_DATA	56 (1 KB)	NTFS Logged ...	22/03/2019 4:34:56 AM	13/03/2019 4:42:18 AM	22/03/2019 4:34:56 AM
	[] \$UpCase	131,072 (128 KB)	Regular File	13/03/2019 4:42:18 AM	13/03/2019 4:42:18 AM	13/03/2019 4:42:18 AM
	[] \$Volume	0 (0 KB)	Regular File	13/03/2019 4:42:18 AM	13/03/2019 4:42:18 AM	13/03/2019 4:42:18 AM
	[] .dropbox.device	56 (1 KB)	Regular File	13/03/2019 4:42:20 AM	13/03/2019 4:42:20 AM	13/03/2019 4:42:20 AM
	[] 7z1900-x64.exe	1,447,178 (1,41...	Regular File	22/03/2019 4:34:27 AM	22/03/2019 4:34:27 AM	22/03/2019 4:34:30 AM
	[] AlpacaCare.docx	53,451 (53 KB)	Regular File	17/03/2019 9:52:20 PM	17/03/2019 9:52:20 PM	17/03/2019 9:52:20 PM
	[] AlpacaCare.docx.FileSlack	3,893 (4 KB)	File Slack			
	[] antimalwaresetup.exe	882,680 (862 KB)	Regular File	22/03/2019 1:08:35 AM	22/03/2019 1:08:35 AM	22/03/2019 1:08:38 AM

10. There was a second partition on the drive. What is the letter assigned to it.

As mention in Q9 above, the second partition is labelled as PacaLady. In the User folder, navigating to \Kare\AppData\Roaming\Microsoft\Office\Recent, we see a few files including shortcuts. One of these files is saved as PacaLady (A), inspecting it further confirms that the letter assigned to the partition is A.

 KarenResume.LNK.FileSlack	01/12/2025 10:59 PM	FILESLACK File	3 KB
 PacaLady (A)	13/03/2019 5:46 AM	Shortcut	1 KB
 Pink floral resume	09/02/2019 10:09 PM	Shortcut	2 KB

11. What is the answer to the question Company's manager asked Karen?

Karen replied **TheCardCriesNoMore** to the manager's question (VGhlQ2FyZENyaWVzTm9Nb3Jl) in one of the emails. After further analysis with Cyberchef, I confirmed that the answer is indeed correct.

Hi Michael,

The answer is TheCardCriesNoMore

-Karen

12. What is the job position offered to Karen? (3 words, 2 spaces in between)

Karen was offered a job as an entry level **Cyber Security Analyst** as stated in one of MS' message to her.

The job position we think you'll be an awesome fit for is an entry level cyber security analysts. We want someone who's willing to learn and don't really care about what you know coming in. We'll be in touch with more information about what this job entails (and the set up involved with getting you payed), but wanted to give you some material to study in the mean time.

13. When was the admin user password last changed?

The SAM hive has information about each user accounts. So for this task, I examined the SAM\Domains\Account\Users folder. As the first account in the device, the admin info is stored in 000001F4 subkey. In value V field, the password information is stored in hex format. Parsing the hex bytes to human readable texts revealed that **2025-03-07 16:41:23 UTC** was the last time the admin changed password.



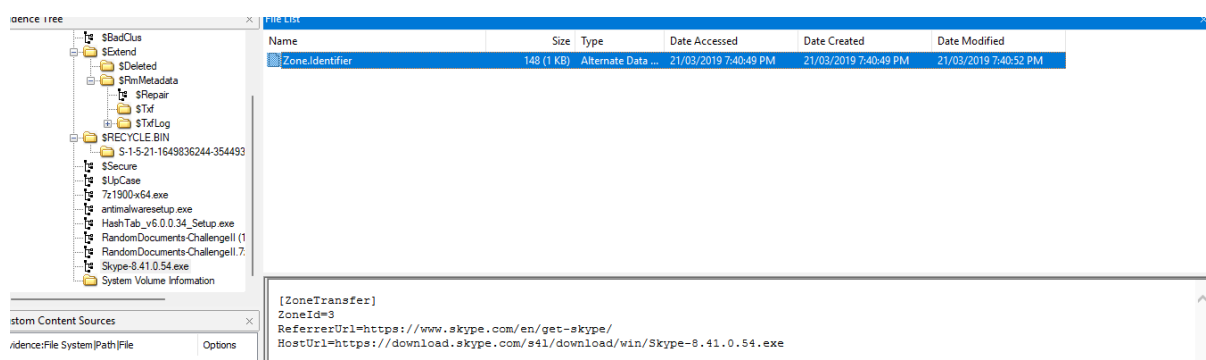
14. What version of Chrome is installed on the machine?

In the SOFTWARE hive, I navigated to \Google\Chrome but there was no useful information. I then explored the WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall where I found the version of installed chrome to be **72.0.2626.121**.

2019-03-13 05:...	Google Chrome	Google Chrome	72.0.3626.121	Google Inc.	20190312		C:\Program Files (x86)\Google\Chrome\Application	"C:\Program Files (x86)\Google\Chrome\Application\72.0.3626.121\Installer\setup.exe" --uninstall --system-level --verbose-loading
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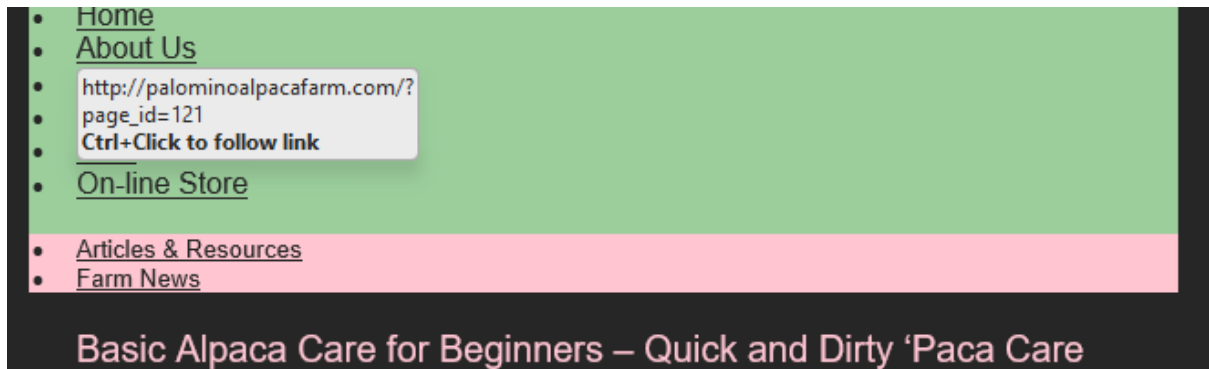
15. What is the URL used to download Skype?

In FTK imager, the root folder mentioned in Q10 has a Skype.exe file in it. Examining the file (Zone.Identifier) that appears after clicking the app file reveals the URL (both host and referred) used in obtaining the app: <https://download.skype.com/s41/download/win/Skype-8.41.0.54.exe>.



16. What is the domain name of the website Karen browsed on Alpaca care that the file AlpacaCare.docx is based on?

The AlpacaCare.docx file had links and write ups about Palomino Alpaca Farm. One of the links directs to <http://palominoalpaca farm.com/>.



Indicators of Compromise

Domain: <http://palominoalpaca farm.com/>

Conclusion

The investigation successfully identified important system and user information, reconstructed user activities and provided indicators of compromise. The findings reveal both normal operational data (installed software, Chrome version etc) and potentially malicious interactions (communication with a hacker group & financial offers). These insights can assist security measures, user monitoring and further forensic or law enforcement actions.